

Examination Scheme for EoSE for Semester-III

CA – Continuous Assessment

EoSE – End of Semester Examination

Regular Students –

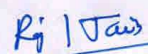
Type of Examination	Course Code and Nomenclature	Duration of Examination		Maximum Marks		Minimum Marks	
Theory	BTH- 63T-351 PLANT BIOCHEMISTRY	CA	01 Hr	CA	20 Marks	CA	08 Marks
		EoSE	03 Hrs	EoSE	80 Marks	EoSE	32 Marks
Practical	BTH-63P-352 PLANT BIOCHEMISTRY-PRACTICAL	CA	1 Hr	CA	10 Marks	CA	04 Marks
		EoSE	04 Hrs	EoSE	40 Marks	EoSE	16 Marks
Theory	BTH-63T-353 ANIMAL BIOCHEMISTRY	CA	01 Hr	CA	20 Marks	CA	08 Marks
		EoSE	03 Hrs	EoSE	80 Marks	EoSE	32 Marks
Practical	BTH-63P-354 ANIMAL BIOCHEMISTRY-PRACTICAL	CA	1 Hr	CA	10 Marks	CA	04 Marks
		EoSE	04 Hrs	EoSE	40 Marks	EoSE	16 Marks
Theory	BTH-63T-355 IMMUNOLOGY	CA	01 Hr	CA	20 Marks	CA	08 Marks
		EoSE	03 Hrs	EoSE	80 Marks	EoSE	32 Marks
Practical	BTH- 63P-356 IMMUNOLOGY -PRACTICAL	CA	1 Hr	CA	10 Marks	CA	04 Marks
		EoSE	04 Hrs	EoSE	40 Marks	EoSE	16 Marks

The theory question paper will consist of **two** parts **A&B**.

PART-A: 20 Marks

Part A will be compulsory having 10 very short answer-type questions (with a limit of 20 words) of two marks each.

PART-B: 60 Marks


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Part B of the question paper shall be divided into four units comprising question numbers 2-5. There will be one question from each unit with internal choice. Each question will carry 15 marks.

UG0804 – Three/Four Year Bachelor of Science (Bio-Technology)								
				Bio-Technology 2 nd YEAR SEM- III			Credits	
#	L e v e l	S e m e s t e r	T y p e	T i t l e			L	T P T o t a l
1.	6	III	MJR	UG0804 – BTH- 63T-351 PLANT BIOCHEMISTRY			4	0 0 4
2.	6	III	MJR	UG0804 – BTH-63P-352 PLANT BIOCHEMISTRY - PRACTICAL			0	0 2 2
3.	6	III	MJR	UG0804 -BTH-63T-353 ANIMAL BIOCHEMISTRY			4	0 0 4
4.	6	III	MJR	UG0804 -BTH- 63P-354 ANIMAL BIOCHEMISTRY -PRACTICAL			0	0 2 2
5.	6	III	MJR	UG0804 -BTH- 63T-355 IMMUNOLOGY			4	0 0 4
6.	6	III	MJR	UG0804 -BTH- 63P-356 IMMUNOLOGY- PRACTICAL			0	0 2 2

BTH- 63T-351 PLANT BIOCHEMISTRY

Semester	Code of the Course	Title of the Course/Paper			NHEQF Level	Credits
III	BTH- 63T-351	PLANT BIOCHEMISTRY			6	4
Level of Course	Type of the Course	Credit Distribution			Offered to NC Student	Course Delivery Method
		Theory	Practical	Total		
Intermediate	Major	4	2	6	NO	60 lectures with diagrammatic and informative assessments during lecture hours
List of Programme Codes in which Offered as Minor Discipline		-----				
Prerequisites		Basic Knowledge of Introductory/Foundation level				

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Objectives of the Course	➤ This module is a general introduction to Photosynthesis, Nitrogen metabolism, plant diseases and plant hormones.
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Course Outcomes:

At the completion of the course, the student would be able to:

Cognitive level	Course outcomes
1. Understanding	➤ The biochemical processes and metabolic pathways, including photosynthesis, photorespiration in plants. ➤ The physiological and biochemical reaction involved in cell wall biosynthesis, nitrogen fixation and assimilation. Illustrate.
2. Memorizing	➤ The biosynthesis, regulation, physiological and biochemical action of plant hormones
3. Applying	➤ The synthesis, physiological and biochemical actions of plant secondary metabolism.

Detailed Syllabus

BTH- 63T-351 PLANT BIOCHEMISTRY

UNIT-I

Photosynthesis: Significance of photosynthesis, Ultrastructure of chloroplast, photosynthesis, Photosynthetic pigments. Light absorption phenomenon, Photosynthesis in C3 and electron transport, Photophosphorylation: Photorespiration, CAM.

Nitrogen metabolism: Metabolism of N- compound in plants, biological nitrogen cycle, nitrogenase structure and function, nitrate reduction, nitrification denitrification, symbiotic and non-symbiotic nitrogen fixation, Nif-gene- organization, function and regulation, Assimilation of fixed nitrogen by plants.

15 Lectures

UNIT-II

Plant hormones: Definition of phytohormones, Auxins, biochemistry and mode of action of auxin, Gibberellin, Cytokinin and other natural growth hormones in plants (ethylene, abscisic acid). Plant stress, Plant responses to abiotic and biotic stresses, Water deficit and drought resistance, Flooding, Temperature stress, Salt stress, Ion toxicity, Pollution stress and potential biotic stress (insects and diseases).

15 Lectures

UNIT-III

Respiration: Overview of glycolysis, Alternative reactions of glycolysis, Regulation of plant glycolysis, Translocation of metabolites across mitochondrial membrane, TCA cycle, electron transport chain, Alternative NAD(P)H oxidative pathways; Cyanide resistant respiration.

15 Lectures

UNIT-IV

Secondary metabolism in plants: Phenolic metabolism shikimate and phenyl propanoid pathways, flavonoids, lignins, and anthocyanins. Isoprenoid metabolism, terpenoids and carotenoids, alkaloids, cyanogenic glycosides and non-protein amino acids. Micro and Macro nutrient deficiency in plants

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BTH-63P-352 PLANT BIOCHEMISTRY-PRACTICAL

1. Photosynthesis related experiments.
2. Chlorophyll estimation by colorimeter.
3. Induction and estimation of hydrolytic enzymes proteinase/amylase/lipase during germination.
4. Vitamin C (Ascorbic Acid) estimation by titration method.
5. Estimation of carotene/ascorbic acid/phenols/tannins in fruits and vegetables.
6. Extraction and assay of urease from Jack bean.

Suggested readings:

1. **Plant Biochemistry:** Hans-Walter Heldt & Heldt, 4th Ed. 2010.
2. **Biochemistry & Molecular Biology of Plant:** Bob B. Buchanan, Wilhelm Gruissem, Russell L. Jones, 2nd Ed. 2015.
3. **Plant Biochemistry:** Dey P. M. Harbone J. B., 1st Ed. 1997.
4. **Advances In Plant Biochemistry:** K.N. P. Singh, Agrotech Press, 2014.
5. **Cell Biology:** Powar C.B., Himalaya Publishing House Mum, 2015.

Course Learning Outcomes:

At the completion of the course, the student would be able to:

1. Understand the significance of photosynthesis, chloroplast structure, photosynthetic pigments, and processes like light absorption, electron transport, photophosphorylation, photorespiration, and CAM pathways.
2. Learn nitrogen metabolism, including the nitrogen cycle, nitrogenase structure, nitrate reduction, nitrification, denitrification, and nitrogen fixation (symbiotic and non-symbiotic), and the assimilation of fixed nitrogen.
3. Comprehend plant hormones such as auxin, gibberellin, cytokinin, ethylene, and abscisic acid, their actions, and their roles in growth and stress responses (abiotic and biotic).
4. Gain knowledge of plant respiration, covering glycolysis, its alternative reactions, regulation, metabolite translocation, the TCA cycle, electron transport, and cyanide-resistant respiration.
5. Explore secondary metabolism, including phenolic pathways, flavonoids, lignins, anthocyanins, isoprenoids, terpenoids, carotenoids, alkaloids, and cyanogenic glycosides.
6. Recognize nutrient deficiencies in plants and understand the biochemical roles of inorganic ions.

BTH-63T-353 ANIMAL BIOCHEMISTRY

Semester	Code of the Course	Title of the Course/Paper		NHEQF Level	Credits
III	BTH-63T-353	ANIMAL BIOCHEMISTRY		6	4
Level of	Type of the	Credit Distribution	Offered	Course Delivery	

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Course	Course	Theory	Practical	Total	to NC Student	Method
Intermediate	Major	4	2	6	NO	60 lectures with diagrammatic and informative assessments during lecture hours
List of Programme Codes in which Offered as Minor Discipline		-----				
Prerequisites		Basic Knowledge of Introductory/Foundation level				
Objectives of the Course		➤ To learn the methodologies for the detection of abnormalities in blood. ➤ To learn the process of different sample collection and processing. ➤ To know about the markers in the various metabolic disorders.				

Course Outcomes:

At the completion of the course, the student would be able to:

Cognitive level	Course outcomes
1. Understanding	➤ The students will be knowing the important biochemical tests ➤ The students will be introduced to methods of processing and analyzing the results.
2. Memorizing	➤ The students will be learning the role of nutrients.
3. Applying	➤ The students will be knowing the procedures and collection and processing of samples.

Detailed Syllabus

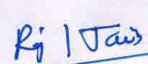
BTH-63T-353 ANIMAL BIOCHEMISTRY

UNIT-I

Scope and importance of biochemistry. Buffer system and pH. Dissociation of and Handerson – Hasselbalch equation Structure of cell membranes and transport across cell membranes. Donnanmembrane equilibrium Biochemistry of carbohydrates: Biological significance (structure and properties) of: Monosaccharides, Disaccharides, Polysaccharides, Mucopolysaccharides.

Metabolism of Carbohydrates: Sequential Reactions and regulation of: Glycolysis, Citric acid cycle and Gluconeogenesis; Phosphate pentose pathway, Glycogenolysis and Glycogenesis; Compartmentalization, Shuttle systems and membrane transporters; Oxidative phosphorylation(Redox system and coupler reaction), Inhibitors and un-couplers of ETC.

15 Lectures


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UNIT-II

Biochemistry of lipids: Properties (structure) and biological significance of simple, compound and derived lipids and lipoproteins Structure and functions of prostaglandins. Chemistry of bile and bile salts; Biochemistry of Lipid: β -oxidation and omega-oxidation of saturated fatty acids(with even and odd number of carbon atoms); Biosynthesis Palmitic acid; Ketogenesis Metabolism of unsaturated fatty acids.

15 Lectures

UNIT-III

Amino acids: Classification and structure of neutral basic and acidic amino acids

Properties of amino acids: amphoteric nature, optical activity and peptide bond formation Structure properties, biological significance and chemical reactions of proteins Degradation of protein: deamination, transamination & decarboxylation of amino acids; Ammonia transport and urea cycle. Protein-protein interactions; Fate of C-skeleton (Glucogenic and Ketogenic amino acids). **15**

Lectures

UNIT-IV

Biochemistry of nucleic acids: Chemistry of purines and pyrimidines, nucleosides and nucleotides. Biological significance of nucleosides and nucleotides.

Structures and functions of deoxyribonucleic acid (DNA) and a typical ribonucleic acid DNA & RNA biosynthesis Integration of metabolism Metabolic functions of macro nutrients Metabolic functions of micro nutrients Metabolic functions of lipid soluble vitamins Metabolic functions of Vitamin A & E.

Metabolic functions of Vitamin D & K Vitamin C, Thiamine, Riboflavin, Niacin & Pyridoxine, Pantothenic acid, Folic acid & Cyanocobalamin. Use of isotope in metabolic studies.

15 Lectures

BTH- 63P-354 ANIMAL BIOCHEMISTRY-PRACTICAL

1. Qualitative and quantitative estimation of Carbohydrates
2. Qualitative and quantitative estimation of Proteins
3. Qualitative and quantitative estimation of Lipids
4. Qualitative and quantitative estimation of Nucleic acid
5. Buffer system and pH
6. Handerson and Hasselbalch equation
7. Titration curve of amino acids
8. Plotting a standard curve using UV-Vis Spectrophotometer and validating the Beer Lambert Law by making BSA protein standard graph by Lowry method

Suggested readings:

1. **Harper's Illustrated Biochemistry**, Robert K Murray et al. (McGraw-Hill Education, 31st Edition)
2. **Principles of Biochemistry**, Voet & Voet (Wiley, 5th Edition)
3. **Principle of Biochemistry**, David L Nelson, Michael M Cox (W.H. Freeman, 8th Edition)
4. **Biochemistry & Molecular Biology of Plants**, Buchanan et al. (American Society of Plant Biologists, 2nd Edition)
5. **Lippincott's Illustrated Reviews- Biochemistry**, Richard Harvey, Denis Ferrier (Lippincott Williams & Wilkins, 7th Edition)
6. **Biochemistry**, Campbell, Farrell (Cengage Learning, 9th Edition)
7. **Biomolecules – Chemistry of Living Systems**, V.K. Ahluwalia (Manakin Press, 2015)
8. **Biochemistry for Medical Students**, Rafi(Orient Blackswan Private Limited - New Delhi, 2014, 2nd Edition)



9. **Textbook of Biochemistry**, S P Singh(CBS publishers and distributors, 2020, 7th Edition)
10. **Biochemistry**, U Satyanarayan(Elsevier, 5th Edition)
11. **Fundamentals of Biochemistry**, J L Jain(S. Chand Publishing, 2004, 7th Edition)
12. **Textbook of Biochemistry for Nurses**, Ashok Kumar J (I. K. International Pvt Ltd)

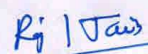
Course Learning Outcomes:

At the completion of the course, the student would be able to:

1. Understand the scope and importance of biochemistry, buffer systems, pH, cell membrane structure, and transport mechanisms.
2. Gain knowledge of carbohydrate biochemistry, including the structure and metabolism of monosaccharides, disaccharides, polysaccharides, and mucopolysaccharides.
3. Comprehend the biochemistry of lipids, including the structure and significance of simple, compound, and derived lipids, as well as lipid metabolism processes like β -oxidation, omega-oxidation, biosynthesis, and ketogenesis.
4. Learn about amino acids and proteins, their classification, structure, properties, and metabolism, including deamination, transamination, and the urea cycle.
5. Understand nucleic acid biochemistry, including the chemistry and biological significance of purines, pyrimidines, nucleosides, and nucleotides, as well as the structure and function of DNA and RNA.
6. Explore the metabolic functions of macronutrients, micronutrients, and various vitamins, and understand the use of isotopes in metabolic studies.

BTH- 63T-355 IMMUNOLOGY

Semester	Code of the Course	Title of the Course/Paper			NHEQF Level	Credits
III	BTH- 63T-355	IMMUNOLOGY			6	4
Level of Course	Type of the Course	Credit Distribution			Offered to NC Student	Course Delivery Method
		Theory	Practical	Total		
Intermediate	Major	4	2	6	NO	60 lectures with diagrammatic and informative assessments during lecture hours
List of Programme Codes in which Offered as Minor Discipline		-----				
Prerequisites		Basic Knowledge of Introductory/Foundation level				


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Objectives of the Course	➤ To articulate the role of various cells and organs involved in immune responses and associated functions ➤ To gain knowledge on the interaction between the immune system and pathogens ➤ To develop the ability to identify issues in clinical immunology.
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Course Outcomes:

At the completion of the course, the student would be able to:

Cognitive level	Course outcomes
1. Understanding	➤ The basic components of immune system and their functions. ➤ Various diagnostic methods based on antigen-antibody interaction ➤
2. Memorizing	➤ Principles and methods of various cellular immune responses
3. Applying	➤ Find effective solutions for the treatment of immune disorders and problem associated.

Detailed Syllabus

BTH- 63T-355 IMMUNOLOGY

UNIT-I

Historical Background-Humoral and Cellular Components of The Immune System, Innate Immunity- Skin And Mucosal Surface, Physiological Barrier, Phagocytic Barrier, Inflammation, Adaptive Immunity, Cells And Organs Of Immune System.

15 Lectures

UNIT-II

Antigens And Antibody-Structure, Properties, Type, Epitopes, Haptens, Antibodies-Structure, Function, Antibody Mediated Functions, Antibody Classes & Biological Activities, Monoclonal Antibodies.
Antigen-Antibody Interaction: Precipitation, Agglutination, RIA, ELISA, Western Blotting.

15 Lectures

UNIT-III

MHC-General Structure and Function, MHC Molecules and Genes, Antigen Processing & Presentation, T Cell Receptor, T Cell Maturation And Differentiation; B Cells-Generation, Activation, Differentiation.

15 Lectures

UNIT-IV

Immune Effector Mechanisms: Cytokines- Properties, Receptors, Antagonist, Secretion; Complement System- Functions, Components, Activation, Regulation and Deficiency, Immune disorders, Cell Mediated Effector Response-Cytotoxic T Cell, Natural Killer Cells, ADCC, Inflammation.

15 Lectures

BTH- 63P-356 IMMUNOLOGY-PRACTICAL

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1. Blood Grouping
2. Differential Leukocyte Count
3. Total Leukocyte Count
4. Widal Test
5. Radial Immunodiffusion (RID)
6. Ouchterlony Double Diffusion
7. Rocket Immuno Electrophoresis
8. Counter Current Immunoelectrophoresis
9. Enzyme Linked Immunosorbent Assay (ELISA) – DOT
10. Immunoprecipitation

Suggested readings:

1. Kuby J. “**Immunology**”, WH Freeman & Co., 5th edition, 2000.
2. M Roitt I., Male., Brostoff, “**Immunology**”, Mosby Publ., 12th edition, 2002.
3. Chakaravarthy A.K., “**Immunology and Immunotechnology**”, Oxford University Press India, 1st Edition, 2006.
4. **The Elements of Immunology** -Fahim Khan.
5. **Immunology**-3rd Edition -Ivan Roitt.

Course Learning Outcomes:

At the completion of the course, the student would be able to:

1. Understand the historical background and components of the immune system, including humoral and cellular aspects, innate immunity, and adaptive immunity.
2. Learn about the structure, properties, and functions of antigens and antibodies, including different antibody classes and biological activities, and techniques for studying antigen-antibody interactions.
3. Comprehend the general structure and function of MHC molecules, antigen processing and presentation, and the maturation and differentiation of T and B cells.
4. Explore immune effector mechanisms, including cytokine properties, complement system functions, activation, and regulation, and various cell-mediated effector responses such as those by cytotoxic T cells and natural killer cells.
5. Gain insights into immune disorders and the roles of inflammation and antibody-dependent cellular cytotoxicity (ADCC) in immune responses.

Examination Scheme for EoSE for Semester-IV

CA – Continuous Assessment

EoSE – End of Semester Examination

Regular Students –

Type of Examination	Course Code and Nomenclature	Duration of Examination	Maximum Marks	Minimum Marks
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